MONASH University

STATISTICS

Session 2: Descriptive Statistics

Descriptive Statistics

- Descriptive statistics are methods used to organise, summarise and present data in a clear and informative way.
- A raw dataset may contain hundreds or thousands of observations, making it difficult to identify important patterns simply by looking at the individual values. Descriptive statistics help transform these values into information that is easier to understand.
- The purpose of descriptive statistics is to describe the data that have been observed. They can help us answer questions such as:
    - What is a typical value?
    - How much do the observations differ from one another?
    - Which values or categories occur most frequently?
    - Are there any unusual observations?
    - What patterns or relationships are visible in the data?

Categorical Data – An example

Suppose we had some data on 5,000 people including information on whether:

1. The individual has been diagnosed with a particular medical condition
   - 'Asthma',
   - 'Cancer',
   - 'Depression',
   - 'Diabetes',
   - 'Heart Disease'
   - 'None of Above'
1. The amount of exercise they do.
   - 'Moderate exercise'
   - 'Minimal exercise'



MONASH University

Categorical Data – An example

- Here is a snapshot of only 8 people in our data.
- Data is arranged in a tabular format, where each row corresponds to a single person.
- For example:
  - Person 1 has no medical conditions and does moderate exercise
  - Person 2 has heart disease and does moderate exercise
  - Etc.

Person ID	Medical Condition	Amount of Exercise
1	None	Moderate
2	Heart Disease	Moderate
3	Diabetes	Moderate
4	None	Moderate
5	None	Minimal
6	None	Moderate
7	Asthma	Minimal
8	None	Moderate

Categorical Data – An example

- When dealing with categorical data, it's common to begin by summarising data with counts (n) and percentages (%)

Person ID	Medical Condition	Amount of Exercise
1	None	Moderate
2	Heart Disease	Moderate
3	Diabetes	Moderate
4	None	Moderate
5	None	Minimal
6	None	Moderate
7	Asthma	Minimal
8	None	Moderate

Condition	Frequency	Percentage
Asthma	1	12.5%
Diabetes	1	12.5%
Heart Disease	1	12.5%
None	5	62.5%

Exercise	Frequency	Percentage
Minimal	2	25%
Moderate	6	75%

Categorical Data – An example

- But we can also look at the variables as pairs (i.e. both at the same time). This is referred to as a contingency table or a cross tabulation.

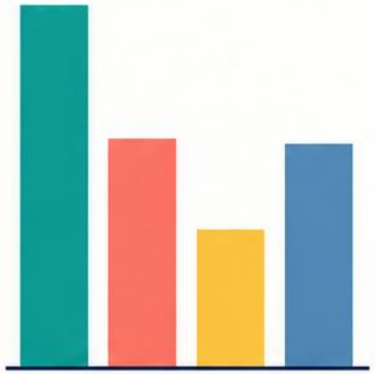
Person ID	Medical Condition	Amount of Exercise
1	None	Moderate
2	Heart Disease	Moderate
3	Diabetes	Moderate
4	None	Moderate
5	None	Minimal
6	None	Moderate
7	Asthma	Minimal
8	None	Moderate

	Exercise		
Condition	Minimal	Moderate	Total
Asthma	1	0	1
Diabetes	0	1	1
Heart Disease	0	1	1
None	1	4	5
Total	2	6	8

Visualising Categorical Data

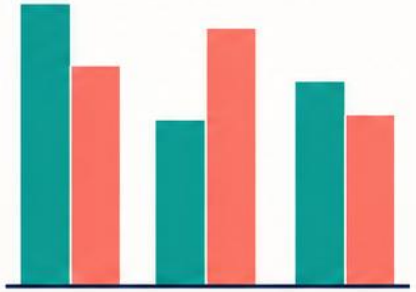
Bar chart

Compare counts or proportions across categories



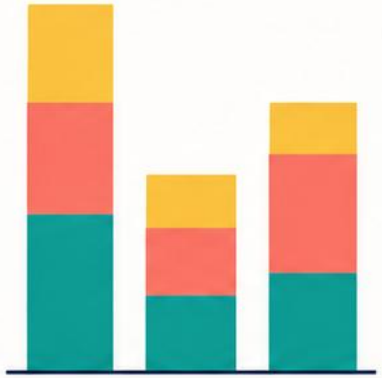
Grouped bar chart

Compare categories across groups



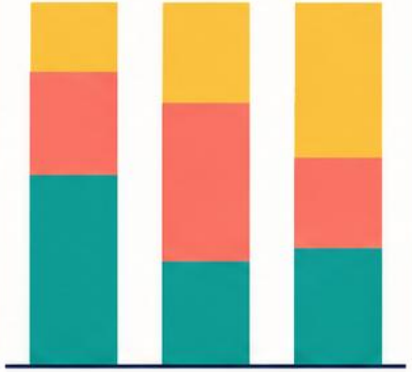
Stacked bar chart

Show the breakdown within each group



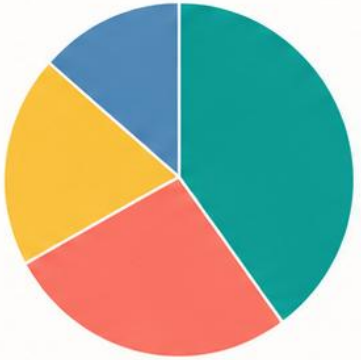
100% stacked bar chart

Compare proportions across groups



Pie chart

Show parts of a whole



Counts show how many • **Proportions help compare groups of different sizes**

MONASH University

Contingency Table and Probabilities

Row Labels	Minimal	Moderate	Total
Asthma	260	166	426
Cancer	256	125	381
Depression	323	170	493
Diabetes	146	56	202
Heart Disease	220	43	263
None of Above	2011	1224	3235
Total	**3216**	**1784**	**5000**

Divide by Row Totals

Row Labels	Minimal	Moderate	Row Total
Asthma	0.6103	0.3897	1
Cancer	0.6719	0.3281	1
Depression	0.6552	0.3448	1
Diabetes	0.7228	0.2772	1
Heart Disease	0.8365	0.1635	1
None of Above	0.6216	0.3784	1
Row Total	**0.6432**	**0.3568**	**1**

Divide by Grand Total

Row Labels	Minimal	Moderate	Grand Total
Asthma	0.0520	0.0332	0.0852
Cancer	0.0512	0.0250	0.0762
Depression	0.0646	0.0340	0.0986
Diabetes	0.0292	0.0112	0.0404
Heart Disease	0.0440	0.0086	0.0526
None of Above	0.4022	0.2448	0.6470
Grand Total	**0.6432**	**0.3568**	**1**

Divide by Column Totals

Row Labels	Minimal	Moderate	Column Total
Asthma	0.0808	0.0930	0.0852
Cancer	0.0796	0.0701	0.0762
Depression	0.1004	0.0953	0.0986
Diabetes	0.0454	0.0314	0.0404
Heart Disease	0.0684	0.0241	0.0526
None of Above	0.6253	0.6861	0.6470
Column Total	**1**	**1**	**1**

Contingency Table and Probabilities

- A contingency table lets us calculate probabilities from counts.
- Marginal probability: use a row or column total. For example, P(Asthma) = 426 / 5000 = 0.0852
- Joint probability: use one cell divided by the grand total. For example, P(Asthma ∩ Moderate) = 166/5000 = .0332
- Conditional probability: use one cell divided by the total for the group named after "given." For example P(Moderate | Asthma) = 166/426 = .3900

Row Labels	Minimal	Moderate	Total
Asthma	260	166	426
Cancer	256	125	381
Depression	323	170	493
Diabetes	146	56	202
Heart Disease	220	43	263
None of Above	2011	1224	3235
Total	**3216**	**1784**	**5000**

Independence (Method 1)

- Two events are independent if the probability of one event occurring is unaffected by the outcome of the other event

Row Labels	Minimal	Moderate	Total
Asthma	260	166	426
Cancer	256	125	381
Depression	323	170	493
Diabetes	146	56	202
Heart Disease	220	43	263
None of Above	2011	1224	3235
Total	**3216**	**1784**	**5000**

Divide by Column Totals

$$\Pr(\text{Diabetes}) = 0.0404$$

$$\Pr(\text{Diabetes}|\text{Minimal}) = 0.0454$$

$$\Pr(\text{Diabetes}|\text{Moderate}) = 0.0314$$

$$\Pr(\text{Diabetes}|\text{Minimal}) \neq \Pr(\text{Diabetes}|\text{Moderate}) \neq \Pr(\text{Diabetes})$$

Row Labels	Minimal	Moderate	Column Total
Asthma	0.0808	0.0930	0.0852
Cancer	0.0796	0.0701	0.0762
Depression	0.1004	0.0953	0.0986
Diabetes	0.0454	0.0314	0.0404
Heart Disease	0.0684	0.0241	0.0526
None of Above	0.6253	0.6861	0.6470
Column Total	**1**	**1**	**1**

Independence (Method 2)

Row Labels	Minimal	Moderate	Total
Asthma	260	166	426
Cancer	256	125	381
Depression	323	170	493
Diabetes	146	56	202
Heart Disease	220	43	263
None of Above	2011	1224	3235
Total	**3216**	**1784**	**5000**

$$\Pr(A \cap B) = \Pr(A) \times \Pr(\mathrm{B})$$

Define:

- A: Person has **Diabetes**
- B: Person does **Minimal** exercise

Obtain probabilities:

- $\Pr(\mathrm{A}) = 202/5000 = 0.0404$
- $\Pr(\mathrm{B}) = 3216/5000 = 0.6432$
- $\Pr(\mathrm{A} \cap \mathrm{B}) = 146/5000 = 0.0292$

Multiply marginal probabilities

- $\Pr(\mathrm{A}) \times \Pr(\mathrm{B}) = 0.0404 \times 0.6432 = 0.0260$

Compare with joint probability

- Does $\Pr(\mathrm{A} \cap \mathrm{B}) = \Pr(\mathrm{A}) \times \Pr(\mathrm{B})$?
- $0.0292 \neq 0.0260$
- $\therefore$ Having diabetes and amount of exercise are not independent

Numerical Data

- When describing numerical data, we typically describe the centre and spread of the data
  - Centre: describes where the values tend to cluster: what is a typical value. Common examples include:
    - The mean (or average): sum of all values divided by the number of values
    - The median: the middle value when the data is ordered
    - The mode: the value that occurs most often in the dataset.

  - Spread: describes how much the values differ from one another. Common examples include:
    - The range: maximum minus the minimum
    - The interquartile range: 75th percentile – 25th percentile
    - The variance: measures how spread out the values are by averaging their squared deviations from the mean.
    - The standard deviation: is the square root of the variance. It describes the spread of values around the mean in the same units as the data.

Measures of Centre: The Mean



$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$= \frac{x_1 + x_2 + \cdots + x_n}{n}$$

$$= \frac{60{,}000 + 59{,}450 + \cdots + 61{,}000}{8}$$

$= 58902.5$ (this is an appropriate 'typical' value for these 8)



- The mean is sensitive to outliers. In this example, the mean would be:

$$\frac{60{,}000 + 59{,}450 + \cdots + 405{,}000{,}000{,}000}{9}$$

$$= 45{,}000{,}052{,}357$$

- This would not tell us what the 'typical' net worth of any of these 9 people

MONASH University

Measures of Centre: The Median

Odd number

Elon Musk	$405,000,000,000
Person 2	$65,200
Person 3	$63,510
Person 4	$61,000
Person 5	$60,000
Person 6	$59,450
Person 7	$55,800
Person 8	$55,060
Person 9	$51,200

$$Median = \left(\frac{n+1}{2}\right)^{th}$$

$$= \left(\frac{9+1}{2}\right)^{th}$$

$$= 5^{th}$$

$$= \$60{,}000$$

Even number

Elon Musk	$405,000,000,000
Person 2	$65,200
Person 3	$63,510
Person 4	$61,000
Person 5	$60,000
Person 6	$59,450
Person 7	$55,800
Person 8	$55,060
Person 9	$51,200
Person 10	$50,500

$$Median = \frac{\left({}^{n}/_{2}\right)^{th} + \left({}^{n}/_{2}+1\right)^{th}}{2}$$

$$= \frac{5^{th} + 6^{th}}{2}$$

$$= \frac{60{,}000 + 59{,}450}{2}$$

$$= \$59{,}725$$

MONASH University

Measures of Centre: The Mode

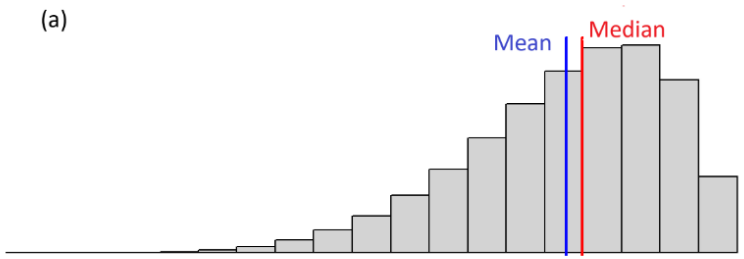
Participant ID	Years of Tertiary Education
1	0
2	0
3	0
4	1
5	1
6	2
7	3
8	3
9	3
10	3
11	3
12	3
13	4
14	4
15	7
16	7
17	10

Participant ID	Social Economic Status
1	Low
2	Low
3	Low
4	Low
5	Medium
6	Medium
7	Medium
8	Medium
9	Medium
10	Medium
11	Medium
12	Medium
13	Medium
14	Medium
15	High
16	High
17	High

MONASH University

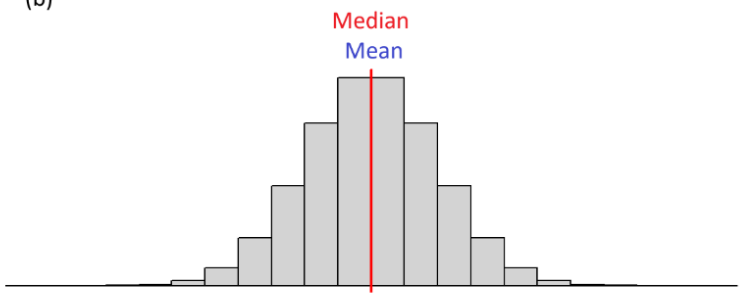
Describing Data Distributions

(a)



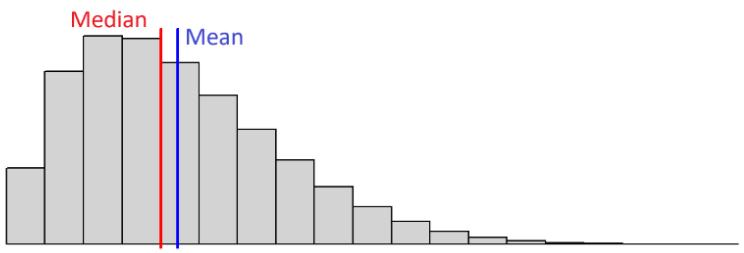
When the mean is lower than the median, the data generally displays negative skewness. This is often due to a few small values in the left tail of the distribution.

(b)



When the mean is approximately equal to the median, the data displays a symmetrical distribution. There is approximately equal number of small and large values the decrease in frequency from the center of the data.

(c)

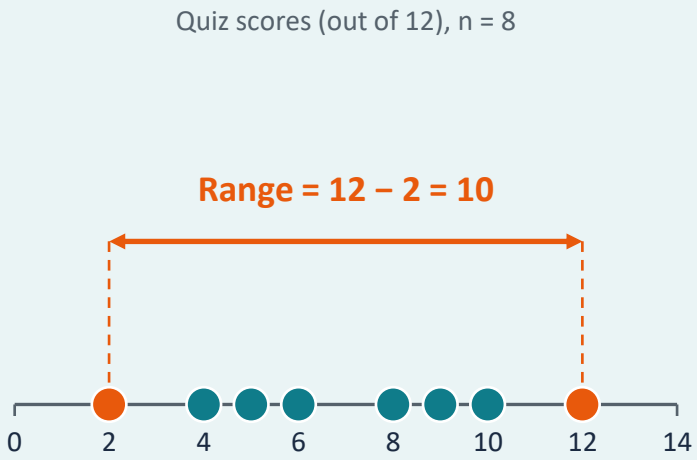


When the mean is larger than the Median, the data generally displays positive skewness. This is often due to a few large values in the right tail of the distribution.

MONASH University

Measures of Spread: The Range

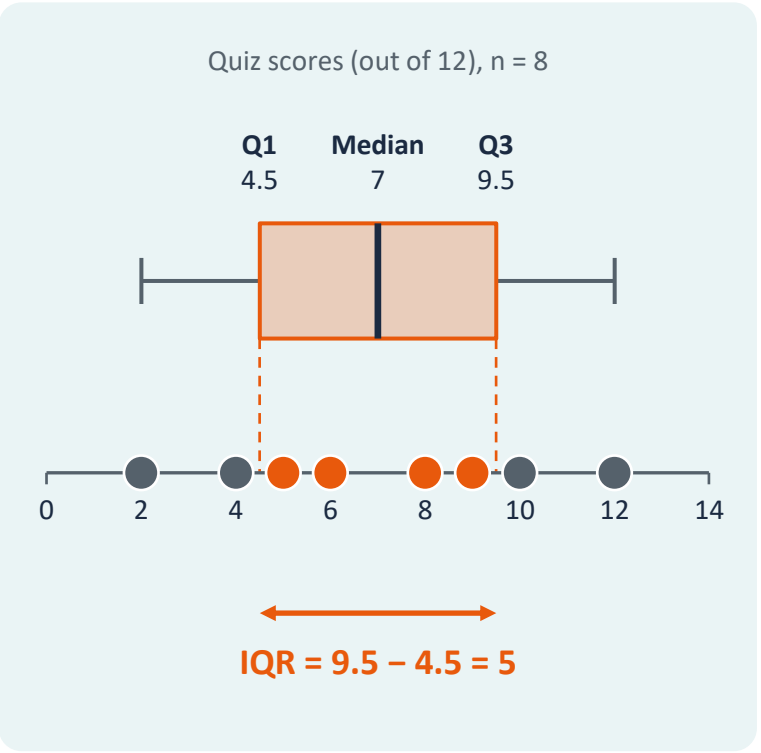
- **Description:** The simplest measure of spread: the distance between the smallest and largest values. It uses only two data points, so a single outlier can change it dramatically.
- **Equation:** **Range = max − min**, where max is the largest observation and min is the smallest.
- **Example:**
  - Quiz scores: 2, 4, 5, 6, 8, 9, 10 and 12, so
  - the range is 12 − 2 = 10 marks.



Only the two orange values (min and max) affect the range; the other six dots are ignored.

Measures of Spread: The Interquartile Range

- **Description:** The spread of the middle 50% of the data, measured between the first and third quartiles. Because it ignores the extreme values, it is resistant to outliers.
- **Equation:** IQR = Q3 − Q1, where Q1 is the 25th percentile (median of the lower half) and Q3 is the 75th percentile (median of the upper half).
- **Example:**
  - Quiz scores: 2, 4, 5, 6, 8, 9, 10 and 12
    - Q1 = (4 + 5) / 2 = 4.5 and
    - Q3 = (9 + 10) / 2 = 9.5, so
    - IQR = 9.5 − 4.5 = 5 marks.



MONASH University

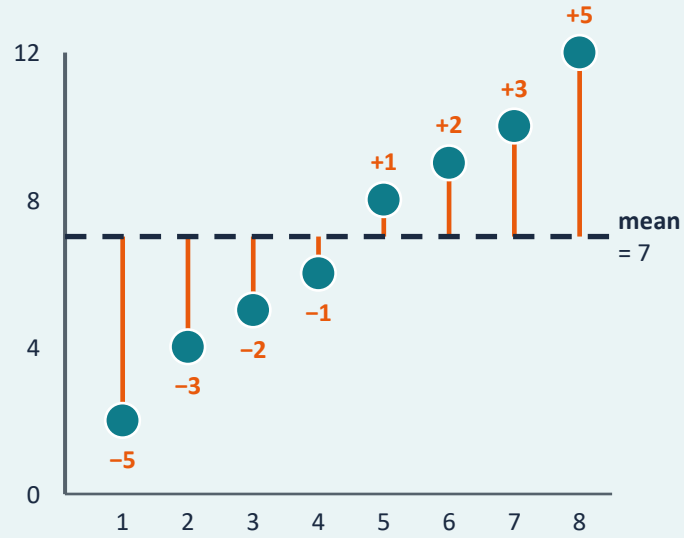
Measures of Spread: The Variance

- **Description:** The average squared distance of each value from the mean. Squaring stops positive and negative deviations cancelling out and gives extra weight to values far from the mean.

- **Equation:** $s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$

- **Example:**
  - Quiz scores: 2, 4, 5, 6, 8, 9, 10 and 12
  - The mean = 7

$$s^2 = \frac{(2-7)^2+(4-7)^2+\cdots+(12-7)^2}{8-1} = 11.14 \text{ marks}^2$$



Each orange line is a deviation from the mean. Variance averages their squares.

MONASH University

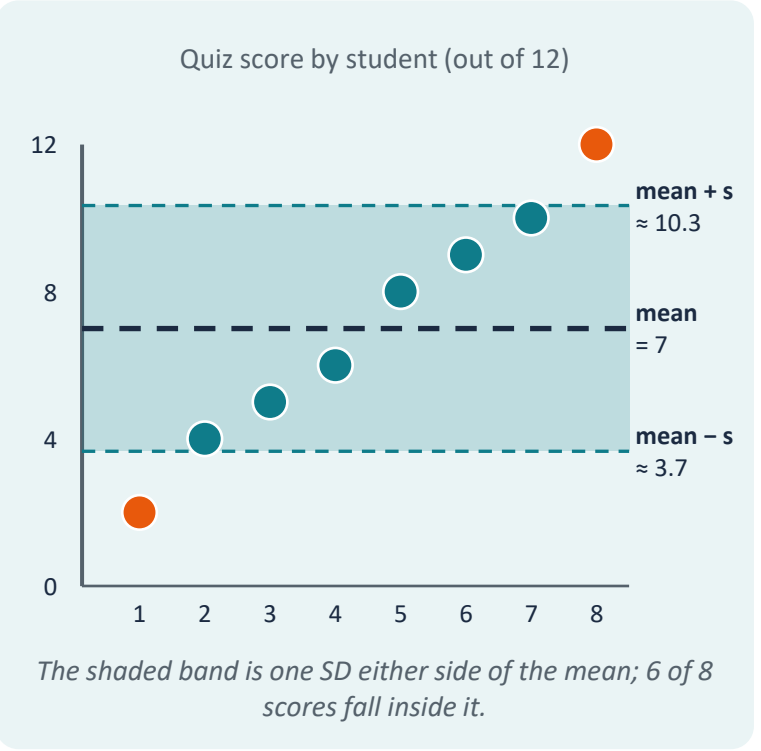
Measures of Spread: The Standard Deviation

- **Description:** The square root of the variance. It returns the spread to the original units of the data, so it describes a typical distance of a value from the mean.

- **Equation:** $s = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$

- **Example:**
  - Quiz scores: 2, 4, 5, 6, 8, 9, 10 and 12
  - The mean = 7

$$s = \sqrt{\frac{(2-7)^2+(4-7)^2+\cdots+(12-7)^2}{8-1}} = 3.3 \text{ marks}$$

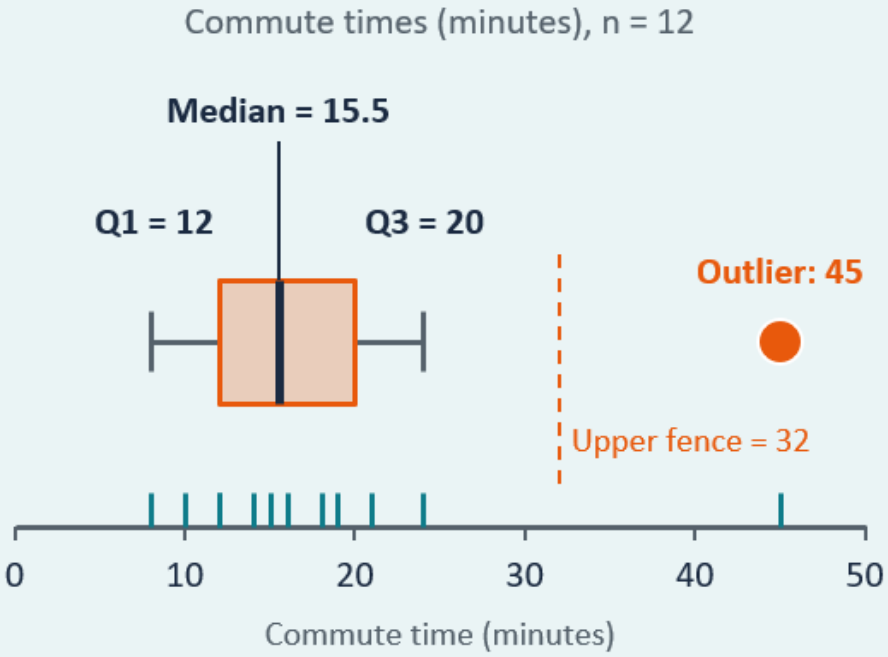


The shaded band is one SD either side of the mean; 6 of 8 scores fall inside it.

MONASH University

Visualising Numerical Data: Boxplots

- **Description:** A boxplot summarises the distribution using five numbers: minimum, Q1, median, Q3 and maximum. The box holds the middle 50% of the data, and points far from the box are flagged as outliers.
- **Rule:** Fences = Q1 − 1.5 × IQR and Q3 + 1.5 × IQR. Whiskers extend to the most extreme values inside the fences; anything beyond them is plotted as a separate point.
- **Example:** Twelve commute times (minutes) give Q1 = 12, median = 15.5 and Q3 = 20, so IQR = 8 and the fences are 0 and 32. The 45-minute trip is an outlier, and the whiskers end at 8 and 24.

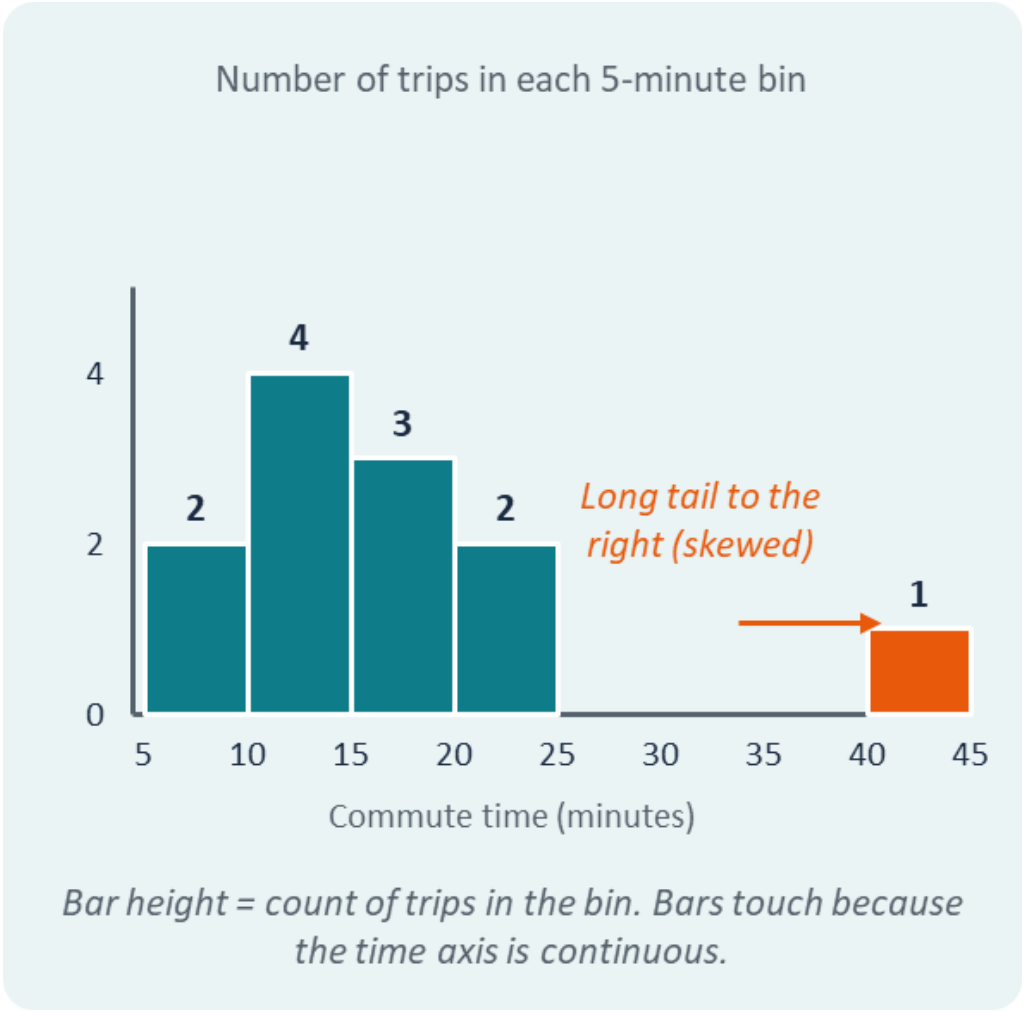


Each teal tick is one trip. Whiskers stop at the last value inside the fences, so 45 is drawn on its own.

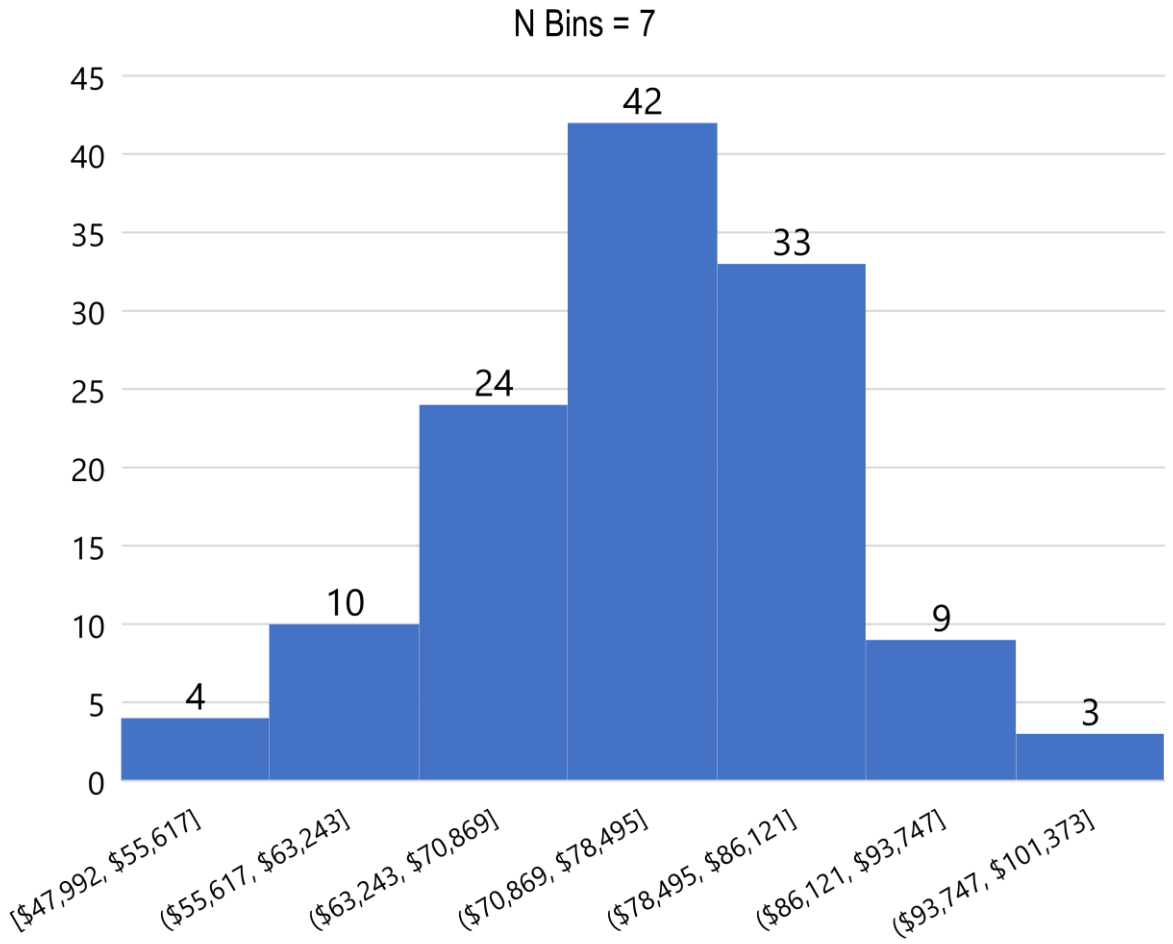
MONASH University

Visualising Numerical Data: Histogram

- **Description:** A histogram splits numerical data into equal-width bins and draws a bar for each one. The bar height is the count of values in that bin, so the shape shows the centre, spread and skewness.
- **Example:** The same commute times with bins (from the boxplot) of width 5 give counts of 2, 4, 3, 2, 0, 0, 0 and 1. The data are right-skewed, with the 45-minute trip alone in the far-right bin.



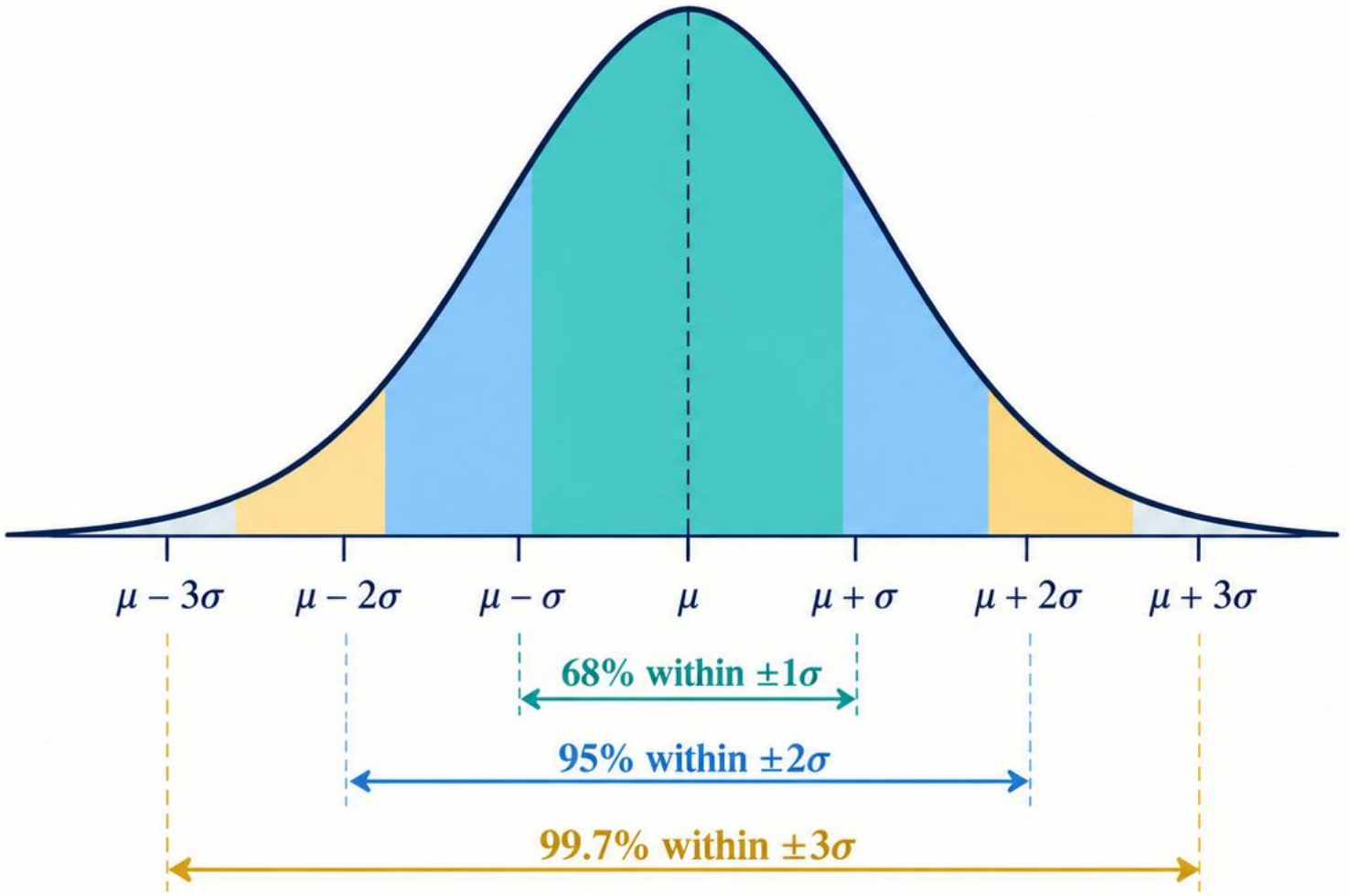
Histograms and Probability Distributions



Income Range	Frequency	Probability
[$47992, $55617]	4	0.032
($55617, $63243]	10	0.080
($63243, $70869]	24	0.192
($70869, $78495]	42	0.336
($78495, $86121]	33	0.264
($86121, $93747]	9	0.072
($93747, $101373]	3	0.024

MONASH University

Properties of the Normal Distribution



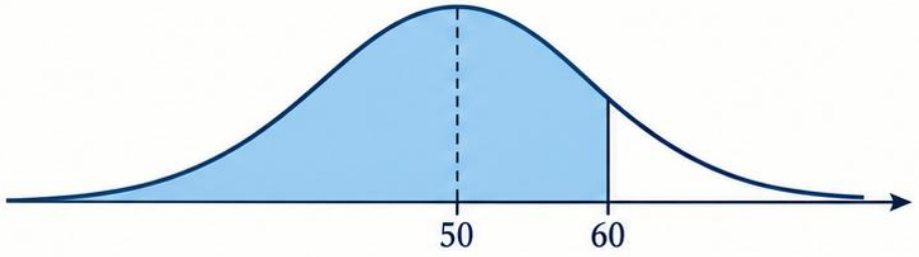
1. Symmetric around μ.
2. Mean = median = mode = μ.
3. Total area under the curve = 1.
4. Tails approach, but never touch, the horizontal axis.
5. μ sets the centre; σ controls the spread.

Common Probability Questions for the Standard Normal Distribution

$X \sim \text{Normal}(\mu = 50, \sigma = 10)$ | **Standardise:** $z = \frac{x - 50}{10}$; $\Phi(z) = P(Z \leq z)$

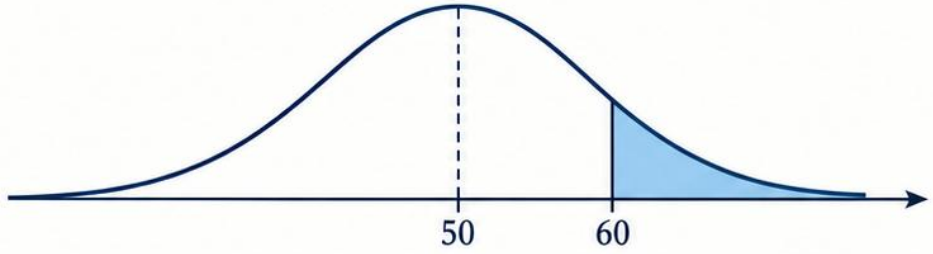
1 Below a value (forward probability)

$$P(X < 60) = \Phi(1) \approx 0.8413$$



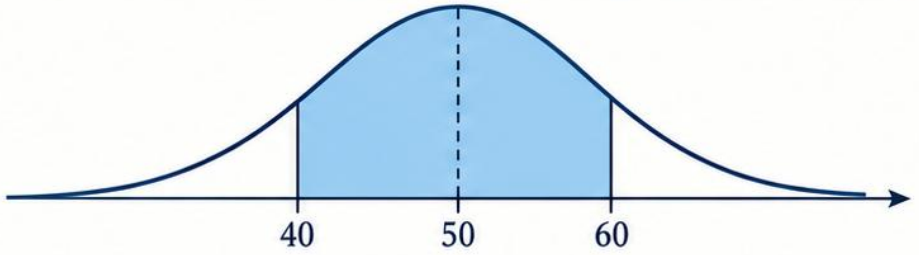
2 Above a value (forward probability)

$$P(X > 60) = 1 - \Phi(1) \approx 0.1587$$



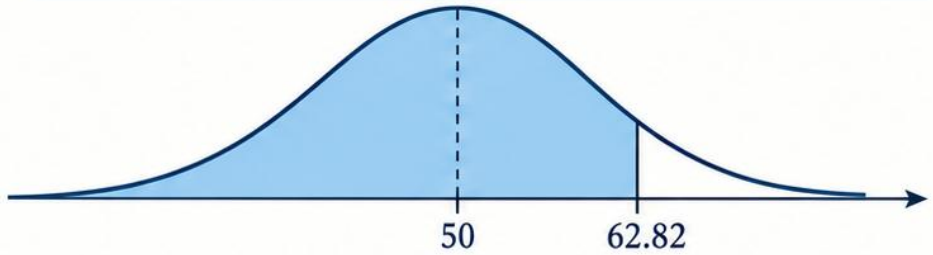
3 Between two values (forward probability)

$$P(40 < X < 60) = \Phi(1) - \Phi(-1) \approx 0.6827$$



4 Area given, find x (inverse percentile)

$$P(X < x) = 0.90 \Rightarrow x = 50 + 10 \times 1.2816 \approx 62.82$$



MONASH University